

# **PREREQUISITE REVIEW: PROBABILITY**

Neural Networks

# PROBABILITY

- Provides a formal way to quantify uncertainty
- Three sources:
  - Stochasticity in the environment
  - Partial observation of the environment
  - Incomplete modelling of the environment

# PROBABILITY INTERPRETATIONS

## **Frequentist probability:**

Analyzing the frequency of events

## **Bayesian probability:**

Modelling degree of certainty/belief

# RANDOM VARIABLE

## **Random Variable:**

A function that maps outcomes in a sample space to real numbers

Example:

Sum of values of two rolled dice

Distance between two particles in an accelerator

# PROBABILITY MASS FUNCTIONS (DISCRETE)



- Indicates the probability a random variable takes on a value  
 $P(X = x)$

Properties:

$$\forall x, 0 \leq P(x) \leq 1$$

$$\sum_x P(x) = 1$$

# PROBABILITY DENSITY FUNCTIONS (CONT)

- Is proportional to the probability of landing inside an infinitely small region at a specific value

Properties:

$$\forall x, p(x) \geq 0$$

$$\int p(x)dx = 1$$

# MARGINAL PROBABILITIES

- A probability distribution computed over a subset of variables

- For discrete variables:

$$\forall x, P(X = x) = \sum_y P(X = x, Y = y)$$

- For continuous variables

$$p(x) = \int p(x, y) dy$$

# CONDITIONAL PROBABILITY

- Describes the probability one event occurs, given another event has occurred

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

$$P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$



# INDEPENDENCE

- Two events or random variables are independent if information about one does not provide information about the other

$$P(A|B) = P(A) \text{ or } P(B|A) = P(B)$$
$$P(A, B) = P(A)P(B)$$

$$\forall x, y \ P(X = x|Y = y) = P(X = x) \text{ or } \forall x, y \ P(Y = y|X = x) = P(Y = y)$$
$$\forall x, y \ P(X = x, Y = y) = P(X = x)P(Y = y)$$

# CONDITIONAL INDEPENDENCE

- Two events or random variables are conditionally independent given another if information about one does not provide information about the other

$$P(A, B|C) = P(A|C)P(B|C)$$

$$\forall x, y \ P(X = x, Y = y|Z = z) = P(X = x|Z = z)P(Y = y|Z = z)$$

# CHAIN RULE OF CONDITIONAL PROBABILITY



- From our definition of conditional probability we get

$$P(A, B) = P(A|B)P(B)$$

$$P(Y = y, X = x) = P(Y = y|X = x)P(X = x)$$

In general, the chain rule of conditional probability is:

$$P(A_1, \dots, A_n) = \prod_{i=1 \dots n} P(A_i | A_1, \dots, A_{i-1})$$

$$P(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1 \dots n} P(X_i = x_i | X_1 = x_1, \dots, X_n = x_n)$$

# EXPECTED VALUE

- The mean value a function of a random variable takes on when the random variable is drawn from a particular distribution

$$E_{X \sim P}[f(x)] = \sum_x P(X = x)f(x)$$

$$E_{x \sim p}[f(x)] = \int p(x)f(x)dx$$

# VARIANCE

- The spread a function of a random variable takes on when the random variable is drawn from a particular distribution

$$\text{Var}[f(x)] = E[(f(x) - E[f(x)])^2]$$

# COVARIANCE

- Estimates the linear relationship between two functions of random variables

$$\text{cov}(f(x), g(y)) = E[(f(x) - E[f(x)])(g(y) - E[g(y)])]$$

# BAYES THEOREM

- Tells us conditional probabilities in terms of other conditional probabilities

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Alternatively, if we think of an event  $E$  as evidence and an event  $H$  as a hypothesis we get

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$